# Analog Circuit-Switched System

A basic analog cellular system consists of three subsystems:

1. A Mobile Unit

2. A Cell Site

3. A Mobile Telephone Switching Office (MTSO) with CONNECTIONS to link the three subsystems

Home | View Diagram

# Basic Cellular Systems

There are mainly two types of Basic Cellular System:

1.Circuit Switched :

In a circuit-switched system, each traffic channel is dedicated to a user until its cell is terminated.

2.Packet Switched:
In packet-switching, the packets are sent towards the destination irrespective of each other. Each packet has to find its own route to the destination. There is no predetermined path; the decision as to which node to hop to in the next step is taken only when a node is reached. Each packet finds its way using the information it carries, such as the source and destination IP addresses.

Analog Circuit-Switched System | Digital Circuit-Switched System | Components of a Basic Cellular System | Packet-Switched System

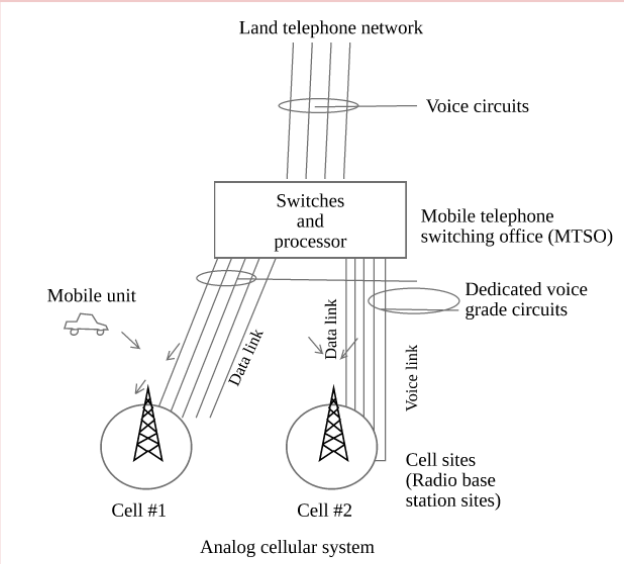
# Digital Circuit-Switched System

A Basic Digital System consists of four elements:

1. Mobile Station (MS)
2. Base Transceiver Station (BTS)
3. Base Station Controller (BSC)
4. Switching Subsystems

Home | View Diagram

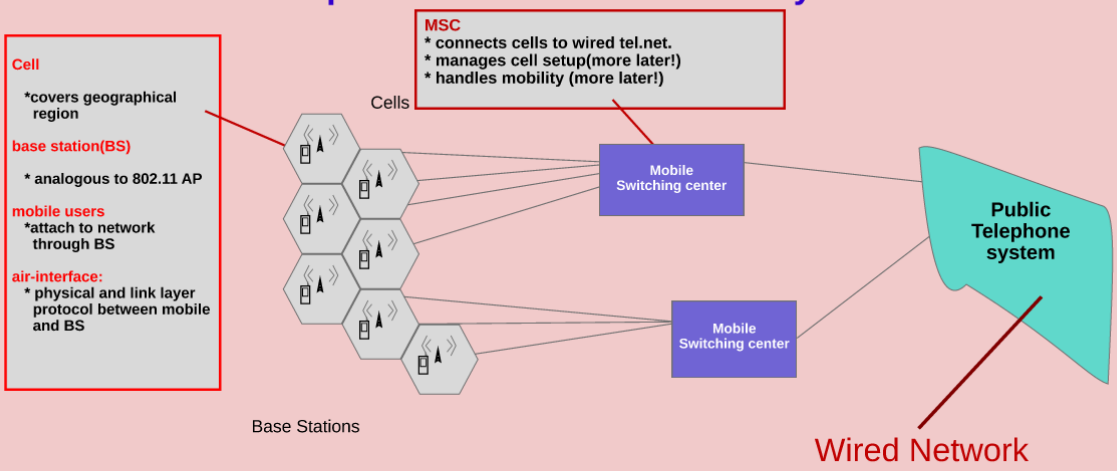
# Analog Circuit-Switched System – Diagram



Analog cellular system

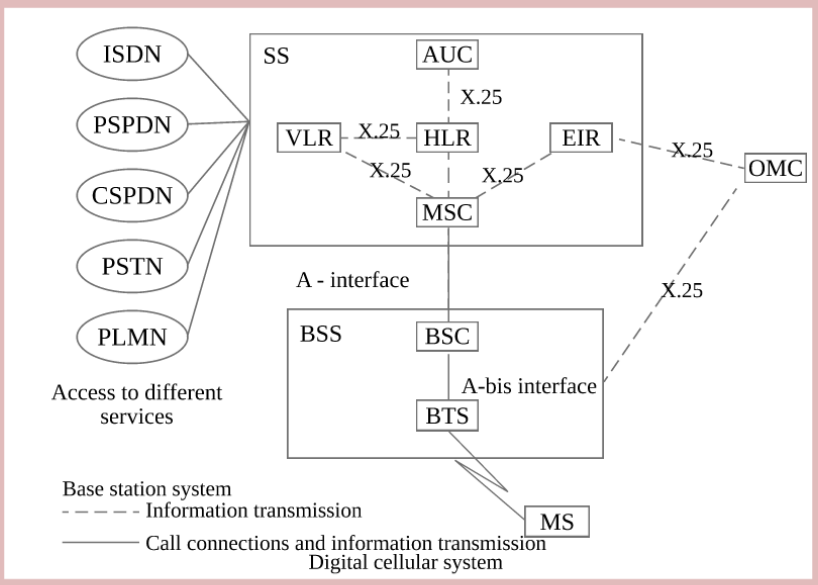
Home | Back

## Components of a Basic Cellular System




Home Back

## Digital Cellular System – Diagram




Home Back

# Packet-Switched System

A cellular packet-switched system has six elements as follows:

**1. MS (Mobile Station)**

**2. Node B**

**3. RNC (Radio Network Controller)**

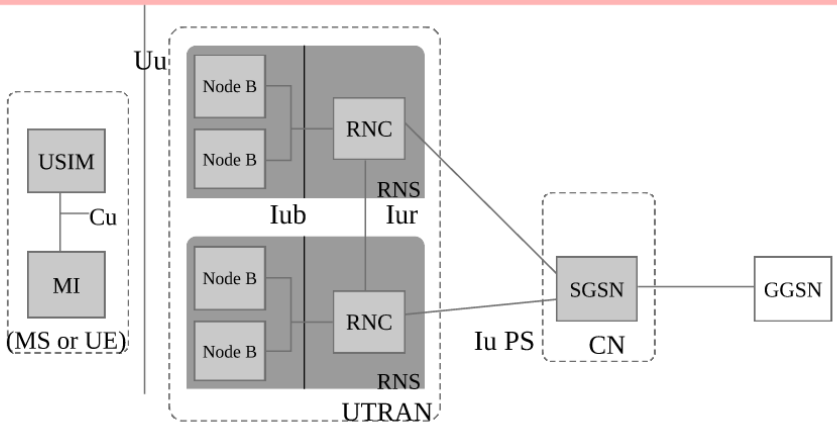
**4. SGSN (Service GPRS Support Node)**

**5. GGSN (Gateway GPRS Support Node)**

**6. CGF (Charging Gateway Function)**

Home | View Diagram

# Cellular Packet System – Diagram




Cellular packet system

Home | Back